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REVIEW 2

Predictive Healthcare Risk StratificationUsingMachine Learning

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Introduction and Problem Statement

Problem: Hospitals and care teams have large amounts of patient data, but it is hard to know early which patientsareat highrisk.Many ML models givea scorebutdo not explainit,do not check if the probability is trustworthy, and may not work equally well for every patient group.

Our idea: build a system that predicts risk from anonymized data, compares several models, explains the result, checks reliability, and gives group-level (cohort) insights.

Literature Survey (1/2)

Ref.	Paper Title	Method	Main Finding	Limitation (Gap)
[1]	Patient risk prediction using historical electronic health records.	Random Forest, XGBoost, SHAP (14 September 2024)	Strong prediction performance across multiple outcomes	Single-hospital dataset
[2]	Explainable machine learning for predicting mortality in sepsis patients	Random Forest, XGBoost, Extra Trees, SHAP (3 June 2025)	Extra Trees achieved AUROC 0.87	Limited to sepsis patients
[3]	The Role of Explainable Artificial Intelligence in Disease Prediction: A Systematic Literature Review and Future Research Directions	Systematic literature review (4 March 2025)	SHAP and LIME improve prediction explanations	Dataset diversity remains a challenge
[4]	RMachine Learning Based Prediction Models for Cardiovascular Disease Risk Using Electronic Health Records Data: Systematic Review and Meta-Analysis	Systematic review and meta-analysis (27 October 2024)	ML models showed promising performance	High variation between studies

Literature Survey (2/2)

Ref.	Paper Title	Method	Main Finding	Limitation (Gap)
[5]	Artificial Intelligence in the Risk Prediction Models of Cardiovascular Disease and Development of an Independent Validation Screening Tool: A Systematic Review	Systematic review and validation assessment	AI models showed potential for cardiovascular prediction	Limited external validation
[6]	Health Risk Assessment Using Machine Learning: Systematic Review	SMOTE oversampliSystematic literature reviewng	ML can support preventive healthcare	No universally applicable model
[7]	Explainable Machine Learning for Predicting ICU Mortality in Myocardial Infarction Patients Using Pseudo-Dynamic Data	Explainable ML for predicting ICU mortality	79 articles and 486 AI models	79 articles and 486 AI models
[8]	Explainable Machine Learning Framework for Dynamic Monitoring of Disease Prognostic Risk: Retrospective Cohort Study	Explainable ML for dynamic disease-risk monitoring	79 articles and 486 AI models	79 articles and 486 AI models

[1] Patient Risk Prediction Using Historical Electronic Health Records

Paper Title: In-hospital Mortality, Readmission, and Prolonged Length of Stay Risk Prediction Leveraging Historical Electronic Patient Records

Authors: Rajeev Bopche et al.

Publication Date: 14 September 2024

Method: Random Forest, XGBoost, traditional machine learning models and SHAP-based feature importance analysis.

Data / Focus: Electronic patient records from a tertiary-care hospital. The study predicts 30-day mortality, hospital readmission and prolonged length of stay.

Main Finding: The models achieved AUROC values of 0.9625 for mortality, 0.8198 for readmission and 0.9797 for prolonged hospital stay.

Limitation (Gap): The study used data from a single hospital. Further validation using different hospitals and patient populations is required.

Paper Link: https://pmc.ncbi.nlm.nih.gov/articles/PMC11401612/

Existing Systems vs Proposed System

Feature	Existing Systems	Proposed System
Risk prediction	Often included	Low / Moderate / High categories with probabilities
Multiple model comparison	Not always included	Logistic Regression, Decision Tree, Random Forest compared
Reasons behind predictions	Not always explained	SHAP and feature importance
Missing data handling	Not always explicit	Clear imputation steps
Imbalanced data handling	Not always explicit	Class weights / SMOTE on training data
Reliability of probabilities	Not always checked	Calibration curves and Brier score
Performance across groups	Not always checked	Subgroup evaluation (age, sex)
Cohort-level analysis	Not always included	Group-level risk dashboard

How Our Project Is Different

One complete pipeline

Prediction, explanation,calibration and subgroup checks are done together, not separately.

Reliable probabilities

We checkthat a 70% riskreally means about 70%, so the numbers can be trusted.

Fair across groups

Wetestresultsfordifferent patient groups and different data conditions.

Group-level insight

Resultsareshownforwhole cohorts, not only single patients.

It supports doctors and analysts. It does not diagnose or replace medical judgment.

Methodology

1
Data Collection
Anonymized healthcare dataset (clinical, demographic, usage data)

2
Preprocessing
Removeduplicates, fix errors, fill missing values, encode and scale

3
Feature Preparation
Selectusefulfeatures;split into train, validation and test sets

4
Imbalance Handling
Classweights orSMOTEon training data only

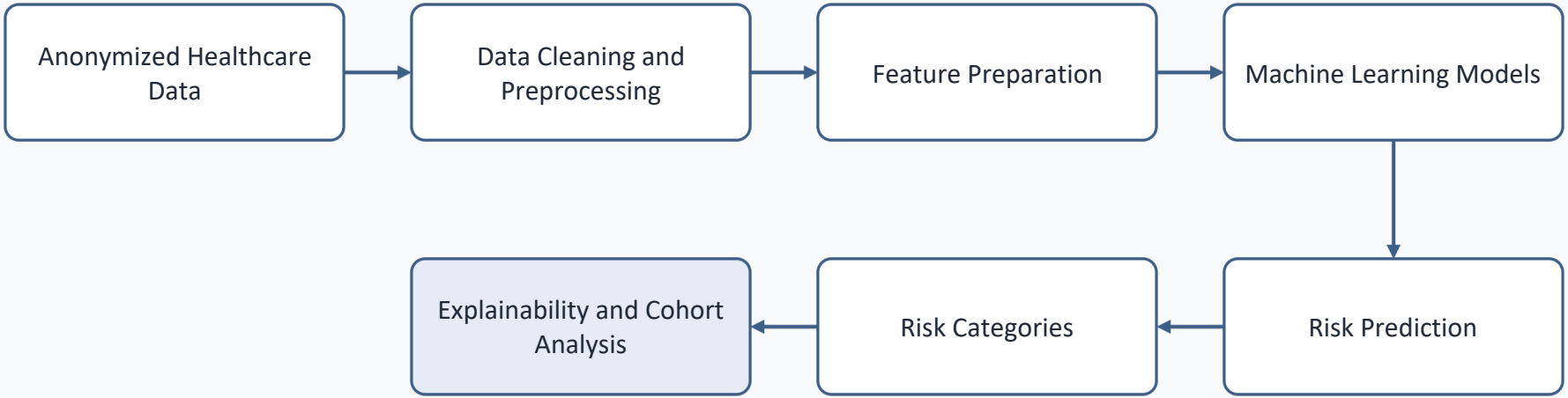
5
Model Training
LogisticRegression, Decision Tree, Random Forest with cross-validation

6
Evaluation
Accuracy,Precision, Recall, F1, ROC-AUC, calibration

7
Explainability
SHAPandfeature importance; subgroup checks

8
Risk Output
Low/Moderate / High categories and cohort analysis

Proposed System Architecture



Low Risk | Moderate Risk | High Risk

Output: risk probability, risk category, key factors and cohort insights. Not an automatic medical decision.

Dataset, Evaluation and Expected Outcome

Dataset and Evaluation

- Public anonymized healthcare dataset (final choice being decided)
  Train / validation / test split
- Metrics: Accuracy, Precision, Recall, F1, ROC-AUC
- Reliability: calibration curve, Brier score
- Subgroup results by age and sex

Expected Outcome

- Comparison of three models on the same data
- Bestmodelchosenusingaccuracyand calibration
- Top risk factors explained with SHAP
- Reliable results across patient groups
- Cohort-level risk report

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Thank You!